Mixed KAN Convolutional Network With Superpixel-Based Multihead Graph Learning for Hyperspectral Image Classification

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***Abstract*—Although convolutional neural networks (CNNs) and graph convolutional networks (GCNs) have advanced hyperspectral image (HSI) classification, each faces its own limitations. Unlike these two models with fixed nonlinear activation functions, Kolmogorov–Arnold networks (KANs) employ learnable activation functions on neuron edges, endowing the network with the ability to adaptively model complex input–output relationships in HSI data. To address these gaps, this article proposes a mixed KAN convolutional network with superpixel-based multihead graph learning (MGL-MKCNet). The MGL-MKCNet framework primarily comprises two branches: the first is a superpixel-based multihead graph (SMHG) branch employing superpixel encoders and decoders for macrofeature learning, while the second is a mixed KAN convolution (MKConv) branch for pixel-level feature extraction. This robust integration enables superior exploitation of the multidimensional features inherent in HSI data. In the pixel-level feature extraction branch, we design a spectral–spatial collaborative attention (SSCA) module, which leverages spatial guidance to extract initial features from spectral information. By integrating a multisemantic spatial attention (MSSA) mechanism with a progressive spectral self-attention (PSSA) strategy, SSCA enables more discriminative and context-aware feature representation. Furthermore, we introduce the MKConv module, which utilizes the characteristics of KANs further to enhance the classification performance. In the superpixel-level feature extraction branch, an SMHG module is designed to overcome the limitations of GCNs in handling dynamic network structures. The full-graph training process is optimized by adopting hierarchical and incremental learning strategies of the superpixel-based graph. Experiments on four HSI datasets demonstrate that the proposed MGL-MKCNet achieves significant improvements in key classification metrics, outperforming multiple state-of-the-art deep learning models.**

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I. Introduction

EACH pixel in a hyperspectral image (HSI) possesses continuous spectral information, fully reflecting the rich reflective characteristics of surface materials across different spectral bands [1]. Unlike conventional RGB imaging systems, hyperspectral imaging systems typically span the range from visible to infrared light, with each band having a bandwidth of less than 10 nm. This unique configuration significantly enhances the ability to distinguish between different materials. Consequently, hyperspectral imaging technology finds extensive application in fields such as crop identification, mineral exploration, and environmental monitoring [2], [3], [4], [5].

Due to its numerous advantages, HSI classification has attracted significant attention from researchers in recent years [6]. In the early stages, various supervised machine learning methods, such as maximum likelihood estimation (MLE) [7] and support vector machines (SVMs) [8], were widely employed for HSI classification. However, these approaches relied heavily on handcrafted feature extraction, making it challenging to automatically capture complex spectral–spatial patterns and limiting their capacity to model spatial contextual information effectively. In recent years, the rapid advancement of deep learning in HSI classification has effectively addressed these limitations. By automatically learning intricate spectral–spatial features and capturing complex nonlinear relationships inherent in the data, deep learning methods have shown great potential in HSI classification and have become a central focus of research in this field [9], [10].

Deep learning methods, including convolutional neural networks (CNNs) [11], Transformers [12], [13], [14], Mamba [15], and graph convolutional networks (GCNs) [16], [17], have been widely used in HSI classification. Among these, CNNs excel at extracting local spatial–spectral features via convolution and hierarchical pooling. Representative works include an automated spectral–spatial CNN framework [18], a binary quantized accelerated CNN [19], a multiscale large-kernel CNN [20], and an unsupervised multiscale feature learning CNN [21]. Despite their strong local modeling ability, CNNs are constrained by regular grid structures and struggle

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to capture non-Euclidean global relationships or long-range pixel dependencies, limiting cross-region and heterogeneous structure learning.

To overcome the limitations of CNNs, the GCNs model HSI data using flexible graph structures, yielding better generalization in small-sample and complex scenes [22]. However, conventional GCNs suffer from three critical drawbacks: full-graph dependency, quadratic computational and storage overhead, and poor scalability when directly using pixels as graph nodes. To alleviate these issues, Wan et al. [23] designed a multiscale dynamic GCN for HSI classification. Liu et al. [24] proposed a CNN-enhanced GCN (CEGCN) that fuses pixel-level and superpixel-level features for HSI interpretation. Liu et al. [25] further developed a fast dynamic graph convolutional network and CNN (FDGC) parallel network for HSI classification. In addition, traditional GCNs use simple weighted feature aggregation and are sensitive to noise. By contrast, the graph attention network adopts an adaptive attention mechanism to dynamically assign learnable weights to nodes, strengthening feature representation without relying on fixed adjacency matrices. Inspired by this, Dong et al. [26] developed the WFCG model, integrating CNNs and graph attention network to jointly exploit local spatial features and global graph-based dependencies for improved HSI classification. Although CNNs and GCNs demonstrate strong performance in HSI analysis, both share the disadvantage of still relying on the structural paradigm of linear transformations with fixed activation functions.

To address the bottleneck of fixed activation functions, the Kolmogorov–Arnold network (KAN) was proposed [27]. Unlike standard networks that fix nonlinearity at node activations, KANs embed learnable spline functions on edges to model flexible input–output mappings. Building on this idea, Zhang and Zhang [28] extended KANs to graph-structured data, proposing a graph KAN (GKAN). The GKAN enhances feature modeling on graphs by incorporating learnable univariate spline functions into the graph convolution process, enabling more flexible representation of node interactions. However, the computational cost of B-spline evaluations in standard KANs poses a significant efficiency bottleneck. To mitigate this, Li [29] proposed FastKAN, which achieved acceleration and simplification of KANs by approximating B-splines with Gaussian radial basis functions (RBFs). By placing nonlinear operations at the edges in KAN-based methods, rather than using fixed activations, these models can capture complex spectral–spatial interactions that traditional CNNs and GCNs may miss, thereby improving the classification performance of HSI classification.

Existing HSI classification algorithms usually improve either pixel-level spectral–spatial modeling or graph-based contextual reasoning, but rarely integrate both within a unified and purpose-driven framework. In particular, CNN-based and CNN–graph hybrid methods are effective at local feature extraction and contextual aggregation, yet they still rely on fixed activation patterns and are therefore limited in modeling complex nonlinear spectral–spatial interactions. In contrast, KAN-based approaches improve nonlinear representation, but they are typically not designed to explicitly exploit scalable superpixel-level graph structures for global context learning. As a result, there remains a need for an end-to-end architecture that jointly captures global graph-contextual dependencies and adaptive nonlinear pixel-level spectral–spatial representations within a single trainable framework. To address these limitations, we propose a mixed KAN convolutional network with superpixel-based multihead graph learning (MGL-MKCNet), a novel dual-branch architecture for HSI classification, as shown in Fig. 1. The first branch employs a superpixel-based multihead graph (SMHG) module that captures both local spatial organization and non-Euclidean interspectral structures through multiple graph views. The second branch designs a spectral–spatial collaborative attention (SSCA) module to explicitly model the synergies between spatial and spectral dimensions, followed by a pixel-level mixed KAN convolution (MKConv) module to capture complex nonlinear spectral–spatial patterns. The features of both branches are then adaptively fused to generate the final classification predictions. The main contributions of this article are as follows.

1) Unlike existing HSI methods that separately emphasize either graph-based context modeling or KAN-based nonlinear feature learning, we propose a unified dual-branch framework, MGL-MKCNet, that jointly performs region-level graph reasoning and pixel-level nonlinear spectral–spatial modeling in an end-to-end manner.
2) Compared with conventional superpixel GCN-based architectures that use single-path neighborhood aggregation, the proposed SMHG introduces multihead graph learning on superpixel nodes, enabling more expressive and scalable contextual dependency modeling while preserving fine-grained spatial consistency through encoder–decoder mapping.
3) We design an MKConv module that is structurally different from prior single-kernel KAN or FastKAN convolutions. Instead of using only one KAN-based convolution at a fixed receptive field, MKConv integrates parallel FastKAN convolutions with 1×1 and 3×3 kernels, followed by feature fusion and residual refinement. This design enables simultaneous modeling of channel-wise nonlinear interactions and local spatial–contextual dependencies, thereby improving multiscale spectral–spatial representation for HSI classification.
4) To address the insufficient interaction between spectral and spatial cues in existing pixel-level models, we design SSCA to explicitly couple multisemantic spatial attention (MSSA) with progressive spectral self-attention (PSSA), thereby enhancing cross-dimensional feature collaboration before nonlinear convolution and final branch fusion.

The remainder of this article is structured as follows. Section II outlines prior fundamental research related to this model. Section III elaborates in detail on our proposed MGL-MKCNet model, providing comprehensive theoretical derivations for its constituent components. Section IV presents and analyzes comprehensive experimental data and discusses the impact of different parameters on the framework. Section V concludes the article and outlines prospects.

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II. Preliminaries

In this section, we provide a concise overview of GCNs and the KAN convolutional operation. GCN excels at modeling non-Euclidean data by propagating information across graph-structured neighborhoods, making them well-suited for capturing spatial relationships in HSIs through superpixel-based graphs. On the other hand, KANs offer a novel paradigm for neural representation by replacing fixed activation functions with learnable spline-based transformations on edges, enabling more flexible and interpretable modeling of complex spectral nonlinearities. The proposed framework strategically integrates graph-based relational reasoning from GCNs and adaptive spectral–spatial feature learning via the MKConv module into a unified architecture.

A. Graph Convolutional Networks

GCNs are a representative class of graph neural networks that perform graph convolution operations to aggregate neighborhood information around each node [30]. Given an undirected graph $G = (\mathcal{V}, \mathcal{E})$ with n nodes, let $\mathbf{A}$ denote its adjacency matrix and $\mathbf{D}$ the degree matrix. The normalized graph Laplacian is defined as follows:

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}} \tag{1}$$

where $\mathbf{I}$ is the identity matrix. Let the eigendecomposition be

$$\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{T} \tag{2}$$

with $\mathbf{U}$ orthonormal (columns are eigenvectors) and $\mathbf{\Lambda} = diag(\mathbf{\Lambda}_1, \ldots, \mathbf{\Lambda}_n)$ be the eigenvalues. Let $\mathbf{\Lambda}_{max}$ be the largest eigenvalue of $\mathbf{\Lambda}$. Therefore, the convolutional operation in the graph G applied to the input $\boldsymbol{s}$ with a spectral filter $g_\theta(\cdot)$ can be expressed as follows:

$$g_\theta * \mathbf{s} = \mathbf{U} g_\theta(\mathbf{\Lambda})\mathbf{U}^{\mathrm{T}}\mathbf{s} \tag{3}$$

where $*$ denotes the graph convolution and $g_\theta(\mathbf{\Lambda})$ is still a diagonal matrix with each item on its diagonal $g_\theta(\mathbf{\Lambda}_i)$. Here, g_θ can be viewed as a learnable convolutional kernel. However, directly learning $g_\theta(\mathbf{\Lambda})$ requires eigendecomposition and is costly. Defferrard et al. [31] proposed ChebNet to solve the problem of high time complexity. Its core idea is to use Chebyshev polynomials $\{T_i(x)\}_{i=1}^{K}$ in the place of the convolution kernel g_θ, that is,

$$g_\theta(\mathbf{\Lambda}) = \sum_{i=0}^{K} \theta_i T_i\left(\tilde{\mathbf{\Lambda}}\right). \tag{4}$$

Here, $\tilde{\mathbf{\Lambda}} = 2/\mathbf{\Lambda}_{max}\mathbf{\Lambda} - \mathbf{I}$ and assuming that all eigenvalues have been scaled to $[-1, 1]$ so that they can be used as input to the Chebyshev polynomial. Substituting (4) into (3), we can get

$$\begin{aligned} g_\theta * \mathbf{s} &= \mathbf{U}\, g_\theta(\mathbf{\Lambda})\, \mathbf{U}^{\mathrm{T}}\mathbf{s} \\ &= \mathbf{U}\left(\sum_{i=0}^{K} \theta_i T_i\left(\tilde{\mathbf{\Lambda}}\right)\right)\mathbf{U}^{\mathrm{T}}\mathbf{s} \\ &= \sum_{i=0}^{K} \theta_i \left(\mathbf{U} T_i\left(\tilde{\mathbf{\Lambda}}\right)\mathbf{U}^{\mathrm{T}}\right)\mathbf{s} \\ &= \sum_{i=0}^{K} \theta_i T_i\left(\tilde{\mathbf{L}}\right)\mathbf{s} \end{aligned} \tag{5}$$

where θ_i is the learnable parameter, and $\tilde{\mathbf{L}} = 2/\lambda_{max}\mathbf{L} - \mathbf{I}$. Kipf and Welling [32] further simplified ChebNet by setting $K = 1$ ($T_0(\tilde{\mathbf{L}}) = \mathbf{I}$ and $T_1(\tilde{\mathbf{L}}) = \tilde{\mathbf{L}}$), and performing a scaling transformation to make $\lambda_{max} = 2$. Thus, (5) can be simplified as follows:

$$g_\theta * \mathbf{s} = \theta_0\,\mathbf{s} + \theta_1\,(\mathbf{L} - \mathbf{I})\,\mathbf{s} = \theta_0\,\mathbf{s} - \theta_1\,\mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\mathbf{s}. \tag{6}$$

To limit the scale of parameters and prevent network overfitting, further assume $\theta_0 = -\theta_1 = \theta$, the expression is renormalized to yield

$$g_\theta * \mathbf{s} = \theta\left(\mathbf{I} + \mathbf{D}^{-\frac{1}{2}}\mathbf{A}\mathbf{D}^{-\frac{1}{2}}\right)\mathbf{s} = \theta\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\right)\mathbf{s} \tag{7}$$

which satisfies the following conditions:

$$\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}, \qquad \tilde{\mathbf{D}} = \sum_{j} \tilde{\mathbf{A}}_{ij}. \tag{8}$$

Thus, for a node-feature matrix $\mathbf{H}^{(l-1)}$ at layer $l-1$ and the learnable weight matrix $\mathbf{W}^{(l)}$, the GCN update is

$$\mathbf{H}^{(l)} = \sigma\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{H}^{(l-1)}\mathbf{W}^{(l)}\right) \tag{9}$$

where $\sigma(\cdot)$ is a nonlinear activation function.

B. Kolmogorov–Arnold Convolutional Network

Unlike traditional MLPs, KANs [33] replace fixed linear weights with learnable spline-based functions, enabling dynamic and adaptive nonlinear transformations. The Kolmogorov–Arnold representation theorem states that any continuous multivariate function can be expressed as follows:

$$f(x_1, \ldots, x_n) = \sum_{q=1}^{2n+1} \Phi_q\left(\sum_{p=1}^{n} \phi_{q,p}\left(x_p\right)\right) \tag{10}$$

where Φ_q and $\phi_{q,p}$ are univariate functions. In KANs, each ϕ can be implemented as a combination of a spline function and a silu activation function, that is,

$$\phi(x) = w_1 \cdot \mathrm{spline}(x) + w_2 \cdot \mathrm{silu}(x). \tag{11}$$

Here

$$\mathrm{spline}(x) = \sum_{i} c_i B_i(x), \quad \mathrm{silu}(x) = \frac{x}{1 + e^{-x}} \tag{12}$$

where c_i are trainable coefficients and $\{B_i(x)\}$ are B-spline basis functions. Let $\boldsymbol{X}$ be an input image. Based on the definition (11), the KAN convolution operation $K_{w\times h}(\cdot)$ with a kernel of size $w \times h$ at position (i, j) is given by

$$\boldsymbol{Y}_{i,j} = K_{w\times h}\left(\boldsymbol{X}_{i,j}\right) = \sum_{q=1}^{w}\sum_{p=1}^{h} \phi_{q,p}\left(\boldsymbol{X}_{i+q,j+p}\right) \tag{13}$$

where $\boldsymbol{Y}$ is the output feature map.

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III. Methodology

This section details our proposed MGL-MKCNet model, comprising three principal modules: the SMHG module, the SSCA module, and the MKConv module. The proposed architecture is designed in a purpose-driven manner rather than by directly stacking heterogeneous modules. Specifically, SMHG is introduced to address the scalability and long-range dependency issues of graph modeling in HSIs through superpixel-level multihead aggregation; SSCA is used to explicitly enhance spectral–spatial interaction before nonlinear feature transformation; MKConv is employed to improve adaptive nonlinear representation and multiscale local modeling at the pixel level; and the final weighted fusion is adopted to combine the complementary strengths of global contextual semantics and local discriminative details.

Second, the core mechanism of MGL-MKCNet can be understood as a global-to-local and coarse-to-fine representation process. The SMHG branch first performs scalable graph reasoning on superpixels to capture long-range structural dependencies and global contextual semantics. In parallel, the pixel-level branch first applies SSCA to strengthen discriminative spectral–spatial interactions and suppress irrelevant responses and then employs MKConv to further refine these enhanced features through adaptive multiscale nonlinear transformation. Finally, the two branches are fused so that global contextual information can complement local fine-grained discrimination, forming a unified mechanism rather than a simple combination of heterogeneous modules.

A. Superpixel-Based Multihead Graph

The HSI is characterized by high spectral dimensionality and rich spatial context, but modeling each pixel as a node in a graph leads to an enormous computational burden due to the large number of nodes. To mitigate this issue while preserving both spectral and spatial information, we employ simple linear iterative clustering (SLIC) [34] to partition the HSI into compact, spectrally homogeneous superpixels in the joint spectral–spatial domain. This approach significantly reduces the number of graph nodes while maintaining local structural integrity.

Let $\mathbf{X} \in \mathbb{R}^{H\times W\times S}$ denote the input HSI with $H \times W$ spatial resolution and S spectral bands. After applying PCA for dimensionality reduction, we obtain a feature map $\mathbf{X}^p \in \mathbb{R}^{H\times W\times B}$, where $B < S$. To enhance feature representation before graph construction, we apply a 1×1 convolution

$$\mathbf{X}' = \mathrm{Conv}_{1\times 1}(\mathbf{X}^p) \tag{14}$$

where $\mathbf{X}' \in \mathbb{R}^{H\times W\times B}$ is the refined feature map with reduced dimensionality and improved nonlinearity.

However, as each superpixel contains a varying number of pixels, they cannot be directly input into a grid-based model for computation. We, therefore, establish an explicit mapping relationship between pixels and superpixels to achieve cross-scale feature alignment and transformation. This yields the association matrix $\mathbf{P} \in \mathbb{R}^{HW\times C}$ between superpixels and pixels

$$\mathbf{P}_{i,j} = \begin{cases} 1, & \text{if } \hat{\mathbf{X}}_i \in \mathcal{S}_j \\ 0, & \text{otherwise} \end{cases} \qquad \hat{\mathbf{X}} = \mathit{Flatten}(\mathbf{X}') \tag{15}$$

where $\hat{\mathbf{X}}_i$ denotes the ith pixel in $\hat{\mathbf{X}}$, the flattened version of $\mathbf{X}'$. $\mathcal{S}_j$ is the jth superpixel and C is the total number of superpixels. The association matrix enables efficient cross-scale feature transformation. Therefore, we define the graph node feature matrix $\mathbf{E} \in \mathbb{R}^{C\times B}$ by aggregating pixel features within each superpixel

$$\mathbf{E} = \mathrm{Encoder}(\mathbf{X}'; \mathbf{P}) = \hat{\mathbf{P}}^T \hat{\mathbf{X}} \tag{16}$$

where $\hat{\mathbf{P}}$ is the normalized form of $\mathbf{P}$. To capture spatial dependencies among superpixels, we construct a graph based on the graph node feature matrix $\mathbf{E} \in \mathbb{R}^{C\times B}$ where the nodes correspond to the superpixels and the edges represent spatial adjacency. Let $\mathbf{A} \in \mathbb{R}^{C\times C}$ be the adjacency matrix defined based on spatial neighborhood relationships, and $\mathbf{D} \in \mathbb{R}^{C\times C}$ be the degree matrix. Following the GCN framework in (9), we add self-loops and normalize:

$$\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}, \quad \tilde{\mathbf{D}}_{ii} = \sum_j \tilde{\mathbf{A}}_{ij}. \tag{17}$$

Furthermore, we design a multihead graph module to enhance the model's expressive power. For each head $i = 1, \ldots, n$, we apply a linear transformation through a learnable weight matrix $\mathbf{W}_i \in \mathbb{R}^{B\times F_h}$, followed by a GCN-style update with information about self-loops

$$\begin{aligned} \mathbf{H}_i &= \sigma_1\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{E}\mathbf{W}_i\right) \\ &= \sigma_1\left(\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{A}\tilde{\mathbf{D}}^{-\frac{1}{2}}\mathbf{E}\mathbf{W}_i + \tilde{\mathbf{D}}^{-1}\mathbf{E}\mathbf{W}_i\right) \end{aligned} \tag{18}$$

where $\sigma_1(\cdot)$ denotes the LeakyReLU function and $\mathbf{H}_i \in \mathbb{R}^{C\times F_h}$. The term $\tilde{\mathbf{D}}^{-1/2}\mathbf{A}\tilde{\mathbf{D}}^{-1/2}\mathbf{E}\mathbf{W}_i$ aggregates neighborhood information, while $\tilde{\mathbf{D}}^{-1}\mathbf{E}\mathbf{W}_i$ retains the self-node feature scaled by its total degree, thus enhancing local representation.

The outputs from all heads are concatenated to form the final multihead representation

$$\mathbf{H} = \mathrm{Concat}\left[\mathbf{H}_i\right]_{i=1}^{n} \tag{19}$$

where $\mathbf{H} \in \mathbb{R}^{C\times(n\cdot F_h)}$ is the aggregated superpixel-level feature vector.

Finally, to recover the pixel-level representation, we use a decoder that redistributes the superpixel features back to their constituent pixels

$$\tilde{\mathbf{X}} = \mathrm{Decoder}(\mathbf{H}; \mathbf{P}) = \mathrm{Reshape}\left(\hat{\mathbf{P}}\mathbf{H}\right) \tag{20}$$

where $\hat{\mathbf{P}}\mathbf{H}$ performs feature expansion from superpixels to pixels and Reshape$(\cdot)$ restores the original spatial dimensions $H \times W$. The resulting $\tilde{\mathbf{X}} \in \mathbb{R}^{H\times W\times(n\cdot F_h)}$ serves as the enhanced feature map for downstream tasks.

This SMHG module, illustrated in Fig. 1(a), effectively balances computational efficiency and modeling capacity by operating on a coarser graph structure while maintaining fine-grained spatial consistency through pixel-superpixel mapping. It leverages GCN-inspired message passing with multihead attention to capture complex spectral–spatial dependencies in HSIs.

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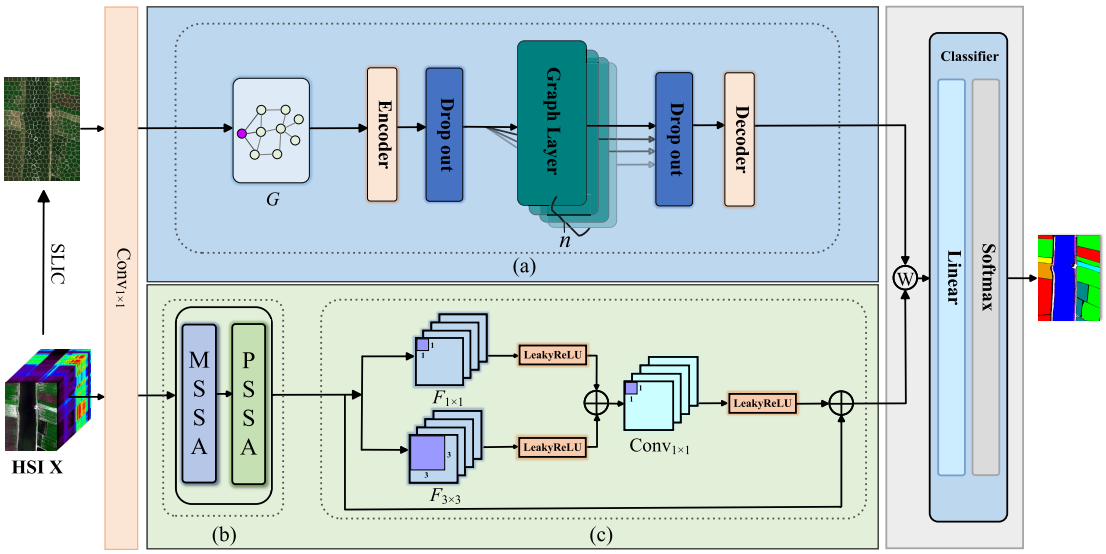


Fig. 1. Overall architecture of MGL-MKCNet. (a) SMHG module. (b) SSCA module, comprising the MSSA and PSSA mechanisms. (c) MKConv module. SLIC segments the HSI into spectrally and spatially similar superpixels, preserving structure while reducing dimensionality. $\text{Conv}_{1\times1}$ denotes a standard 1×1 convolution. $F_{1\times1}$ and $F_{3\times3}$ represent FastKAN convolutions with kernel sizes 1×1 and 3×3, respectively. $\oplus$ indicates pixel-wise addition and (W) denotes weighted summation.

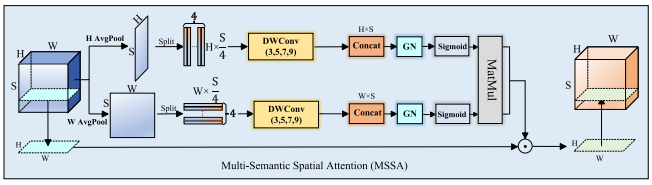


Fig. 2. MSSA mechanism: 1) GN denotes group normalization, 2) $\odot$ denotes element-wise multiplication, and 3) DWConv denotes depthwise convolutions, and (3, 5, 7, 9) denotes the respective sizes of the corresponding convolutional kernels.

B. Spectral–Spatial Collaborative Attention

Traditional CNNs commonly suffer from the problems that spectral and spatial information cannot be fully fused in HSI classification, as well as the serious conflict of multisemantic spatial features. Therefore, inspired by Si et al. [35], we propose an SSCA module. The overall module comprises two components: the MSSA mechanism and the PSSA mechanism. Similarly, in this branch, we first introduce a 1×1 convolution to introduce nonlinearity, thereby enhancing the feature representation capability. Let the input feature map be $\mathbf{X} \in \mathbb{R}^{H\times W\times S}$, we obtain

$$\mathbf{X}^* = \text{Conv}_{1\times1}(\mathbf{X}). \tag{21}$$

1) MSSA: We utilize MSSA to extract salient regions from the features at different spatial scales. The specific process is shown in Fig. 2. First, average pooling (AvgPool) is performed along both spatial dimensions

$$\begin{aligned}\mathbf{X}_H &= \text{AvgPool}_H(\mathbf{X}^*) \in \mathbb{R}^{H\times S}\\ \mathbf{X}_W &= \text{AvgPool}_W(\mathbf{X}^*) \in \mathbb{R}^{W\times S}.\end{aligned} \tag{22}$$

Next, we divide $\mathbf{X}_H$ and $\mathbf{X}_W$ into four spectral subsets

$$\mathbf{X}_H^i = \mathbf{X}_H\left[:, \tfrac{iS}{4} : \tfrac{(i+1)S}{4}, :\right] \qquad \mathbf{X}_W^i = \mathbf{X}_W\left[:, \tfrac{iS}{4} : \tfrac{(i+1)S}{4}, :\right] \qquad i = 1,2,3,4. \tag{23}$$

Each subfeature is subsequently processed independently via a depthwise convolution (DWConv) [36], employing four convolutional kernels of differing sizes, their dimensions being $k_i \in \{3,5,7,9\}$

$$\begin{aligned}\hat{\mathbf{X}}_H^i &= \text{DWConv}_{k_i\times k_i}\left(\mathbf{X}_H^i\right)\\ \hat{\mathbf{X}}_W^i &= \text{DWConv}_{k_i\times k_i}\left(\mathbf{X}_W^i\right).\end{aligned} \tag{24}$$

The subfeatures are concatenated along the spectral dimension, group normalization (GN), and passed through a Sigmoid activation to generate directional attention weights

$$\begin{aligned}\mathbf{A}_H &= \sigma_2\left(\text{GN}\left(Concat\left(\hat{\mathbf{X}}_H^1, \hat{\mathbf{X}}_H^2, \hat{\mathbf{X}}_H^3, \hat{\mathbf{X}}_H^4\right)\right)\right)\\ \mathbf{A}_W &= \sigma_2\left(\text{GN}\left(Concat\left(\hat{\mathbf{X}}_W^1, \hat{\mathbf{X}}_W^2, \hat{\mathbf{X}}_W^3, \hat{\mathbf{X}}_W^4\right)\right)\right).\end{aligned} \tag{25}$$

Finally, the spatial attention is remapped to 2-D space and applied to the original feature map

$$\mathbf{X}_s(:,:,k) = \left(\mathbf{A}_H \times \mathbf{A}_W^T\right) \odot \mathbf{X}^*(:,:,k). \tag{26}$$

Here, $\mathbf{A}_H \in \mathbb{R}^{H\times S}$ and $\mathbf{A}_W \in \mathbb{R}^{W\times S}$ are obtained by reshaping the 1-D attention responses along the width and height axes, respectively, and then broadcast to match the spatial dimensions of $\mathbf{X}^*$. $\mathbf{X}_s$ is the spatially enhanced feature map, which preserves multiscale spatial context information, $\odot$ denotes element-wise multiplication.

2) PSSA: Next, we feed the spatially enhanced features $\mathbf{X}_s$, the final output of the MSSA module, into the PSSA module. The specific process is shown in Fig. 3. This module performs (AvgPool) along the spatial dimension to reduce computational costs, with a focus on spectral modeling. Training is stabilized through GN, yielding

$$\mathbf{X}_1 = \text{GN}\left(\text{AvgPool}_{H\times W}(\mathbf{X}_s)\right) \in \mathbb{R}^{H'\times W'\times S}. \tag{27}$$

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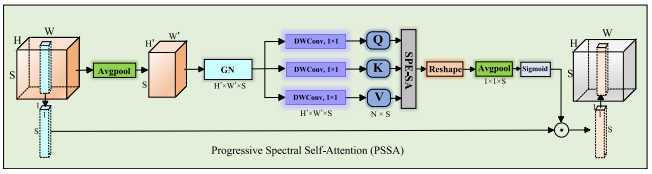


Fig. 3. PSSA mechanism: 1) SPE-SA denotes spectral self-attention, 2) GN denotes group normalization, 3) ⊙ denotes element-wise multiplication, and 4) DWConv denotes depthwise convolutions, 1 × 1 denotes a convolution kernel of size 1. Here, $N = H' \times W'$.

Subsequently, the spectral self-attention (SPE-SA) is computed using query, key, and value projections derived from the input spectral features. Specifically, the query, key, and value were all generated via the DWConv with 1 × 1 kernel as follows:

$$\mathbf{Q}, \mathbf{K}, \mathbf{V} = \mathrm{DWConv}_{1\times 1}(\mathbf{X}_1). \tag{28}$$

The attention distribution and spectral context fusion are computed as follows:

$$\begin{aligned} \mathbf{X}_2 &= \mathrm{Reshape}(\mathbf{A}_s \times \mathbf{V}) \\ s.t. \quad \mathbf{A}_s &= \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right). \end{aligned} \tag{29}$$

Here, $\mathbf{A}_s$ is the resulting SPE-SA matrix for the input $\mathbf{X}_1$. Then, through the AvgPool operation and the sigmoid function, the following is obtained:

$$\mathbf{M} = \sigma_2\left(\mathrm{AvgPool}(\mathbf{X}_2)\right) \in \mathbb{R}^{1\times 1\times S}. \tag{30}$$

Finally, the matrix $\mathbf{M}$ processed through the PSSA mechanism is aggregated with the original input $\mathbf{X}_s$ to yield

$$\mathbf{X}_c(i, j, :) = \mathbf{X}_s(i, j, :) \odot \mathbf{M}. \tag{31}$$

C. Mixed KAN Convolution

In related theoretical work, KAN convolution has shown strong potential for HSI classification due to its adaptive nonlinear representation capability. However, standard KAN convolution suffers from the limitations of B-spline parameterization, including additional spline-grid rescaling and the risk that input variables may exceed the predefined spline domain during training. To alleviate these issues, we draw inspiration from FastKAN [29] and adopt FastKAN convolutions as the basic nonlinear operators for pixel-level feature extraction. FastKAN approximates the original B-spline functions using Gaussian RBFs, thereby improving computational efficiency and training stability.

It should be mentioned that the proposed MKConv is not just a straightforward adaptation of standard FastKAN convolutions. Existing KAN-based convolutions are typically implemented with a single kernel size and thus operate at a fixed receptive field. In contrast, MKConv is designed as a composite module that integrates parallel FastKAN convolutions with complementary kernel sizes, followed by feature fusion and residual refinement. Specifically, the 1 × 1 FastKAN branch focuses on nonlinear channel-wise spectral interaction modeling, while the 3 × 3 FastKAN branch captures local spatial–contextual dependencies. Their outputs are then fused to obtain a richer multiscale nonlinear spectral–spatial representation, and a residual connection is further introduced to preserve the original input information and stabilize optimization. Therefore, the term "mixed" in MKConv refers to the integration of multikernel FastKAN branches and residual feature fusion within a unified module, which makes it structurally different from prior single-path KAN-based convolutions. Among the various RBF options available in FastKAN, the Gaussian function is one of the most commonly used. Therefore, an RBF network with N centers is defined as follows:

$$\varphi(x) = \sum_{i=1}^{N} w_i\, e^{-\frac{x^2}{2h^2}} \tag{32}$$

where w_i is the adjustable weight and h is a bandwidth parameter controlling the width of the function. By aligning the Gaussian RBF with the B-spline through a linear transformation, FastKAN provides an efficient and stable alternative. Based on the FastKAN, we design an MKConv module for pixel-level feature enhancement, with the specific workflow as follows. The subsequent input feature map is $\mathbf{X}$ following output from the SSCA module $\mathbf{X}_c \in \mathbb{R}^{H\times W\times S}$, and the output feature map be $\mathbf{Z} \in \mathbb{R}^{H\times W\times S}$. The computations within the module are defined as follows:

$$\mathbf{Z}_1 = \sigma_1\left(F_{1\times 1}(\mathbf{X}_c)\right) + \sigma_1\left(F_{3\times 3}(\mathbf{X}_c)\right) \tag{33}$$

$$\mathbf{Z} = \sigma_1\left(\mathrm{Conv}_{1\times 1}(\mathbf{Z}_1)\right) + \mathbf{X}_c. \tag{34}$$

Here, $\sigma_1(\cdot)$ denotes the LeakyReLU activation function. $F_{m\times m}(\cdot)$ denotes the FastKAN convolution operation with a kernel of size $m \times m$. Similar to (13), by substituting φ for ϕ, it can be defined as follows:

$$\tilde{\mathbf{Y}}_{i,j} = F_{m\times m}\left(\mathbf{X}_{i,j}\right) = \sum_{q=1}^{m}\sum_{p=1}^{m} \varphi_{q,p}\left(\mathbf{X}_{i+q,j+p}\right). \tag{35}$$

The advantage of combining 1 × 1 and 3 × 3 FastKAN convolutions is that they provide complementary nonlinear modeling capabilities at different receptive fields. The 1 × 1 branch focuses on nonlinear channel mixing and band-wise interaction modeling, which is important for discriminating subtle spectral differences. The 3 × 3 branch captures local spatial structures and neighborhood correlations, which are crucial for preserving spatial continuity and contextual consistency. Their fused representation therefore provides richer spectral–spatial features than either branch alone. In addition, the residual connection helps retain the original input information and improves optimization stability, especially under limited training samples.

D. Feature Fusion and Classification

The two branches within the MGL-MKCNet model employ distinct feature modeling mechanisms, yielding output features that exhibit significant semantic complementarity. Specifically, the graph-based branch primarily characterizes global topological relationships and contextual dependencies between superpixels, whilst the pixel-based branch focuses on extracting spectral–spatial synergistic features and fine-grained discriminative information within local regions. To effectively

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TABLE I
LAND-COVER CLASSES AND SAMPLE DIVISIONS IN FOUR HSI DATASETS

HSIs	Indian Pines (IP)				Pavia University (PU)				WHU-Hi-LongKou (LK)				Augsburg (AG)			
Size	145 × 145 × 200				610 × 340 × 103				550 × 400 × 270				332 × 485 × 180			
No.	Class	Train	Val	Test	Class	Train	Val	Test	Class	Train	Val	Test	Class	Train	Val	Test
1	Alfalfa	1	1	44	Asphalt	7	7	6617	Corn	35	35	34441	Forest	136	136	13235
2	Corn-notill	15	15	1398	Meadows	19	19	18611	Cotton	9	9	8356	Residential area	304	304	29721
3	Corn-min	9	9	812	Gravel	3	3	2093	Sesame	4	4	3023	Industrial area	39	39	3773
4	Corn	3	3	231	Trees	4	4	3056	Broad-leaf soybean	64	64	63084	Low plants	269	269	26319
5	Grass-pasture	5	5	473	Metal Sheets	2	2	1341	Narrow-leaf soybean	5	5	4141	Allotment	6	6	563
6	Grass-trees	8	8	714	Bare Soil	6	6	5017	Rice	12	12	11830	Commercial area	17	17	1611
7	Grass-pasture-mowed	1	1	26	Bitumen	2	2	1326	Water	68	68	66920	Water	16	16	1498
8	Hay-windrowed	5	5	468	Bricks	4	4	3674	Roads and houses	8	8	7108				
9	Oats	1	1	18	Shadow	1	1	945	Mixed weed	6	6	5217				
10	Soybean-notill	10	10	952												
11	Soybean-mintill	25	25	2405												
12	Soybean-clearn	6	6	581												
13	Wheat	3	3	199												
14	Woods	13	13	1239												
15	Buildings-Grass-Trees	4	4	378												
16	Stone-Steel-Towers	1	1	91												
Total		110	110	10029		48	48	42680		211	211	204120		787	787	76720

integrate complementary information from different branches, the model ultimately employs a weighted feature fusion strategy. We know that the output feature of the superpixel-based branch is denoted as $\tilde{\mathbf{X}}$, while that of the pixel-based branch is denoted as $\mathbf{Z}$. The final fusion feature map $\mathbf{F}_s$ is computed as follows:

$$\mathbf{F}_s = \tilde{\mathbf{X}} \textcircled{w} \mathbf{Z} = \eta \cdot \tilde{\mathbf{X}} + (1 - \eta) \cdot \mathbf{Z} \tag{36}$$

where $\eta \in [0, 1]$ is a tunable fusion coefficient that controls the relative contribution of the two branches. After feature fusion, the integrated representation $\mathbf{F}_s$ is fed into a linear classifier followed by a Softmax function to produce the final prediction

$$\hat{\mathbf{Y}} = \text{Softmax}\left(\mathbf{W}_c \mathbf{F}_s + \mathbf{b}_c\right) \tag{37}$$

where $\mathbf{W}_c$ and $\mathbf{b}_c$ denote learnable classification parameters.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

This section presents a comprehensive evaluation of the proposed MGL-MKCNet, including details on the hyperspectral datasets used, experimental settings, comparative methodologies, quantitative results, and qualitative visualizations. To rigorously assess the performance of MGL-MKCNet, we selected eight state-of-the-art HSI classification methods for comparison. All experiments were conducted under identical conditions, with each method configured using its optimal hyperparameters as reported in the original publications to ensure a fair comparison.

The competing models include: CEGCN [24], weighted feature fusion of convolutional neural network and graph attention network (WFCG) [26], fast dynamic graph convolutional network and CNN (FDGC) parallel network [25], spectral–spatial feature tokenization Transformer (SSFTT) [13], dual-view spectral and global spatial feature fusion network (DSGSF) [37], groupwise separable convolutional ViT (GSC-ViT) [14], spatial–spectral Mamba for HSI classification (MambaHSI) [15], and subpixel spectral variability network (S2VNet) [38].

A. Dataset Description and Experimental Setting

To comprehensively evaluate model performance across diverse scenarios, we employed four widely used hyperspectral benchmark datasets: Indian Pines (IP), Pavia University (PU), WHU-Hi-LongKou (LK), and Augsburg (AG). A detailed description of each dataset is provided in Table I.

All experiments reported in this article were implemented using Python 3.12.3 and PyTorch 2.5.1 and conducted on a workstation equipped with an NVIDIA RTX 4060 Ti GPU (16 GB memory) and an Intel Core i5-13600KF CPU. For reproducibility, the key structural hyperparameters of MGL-MKCNet are summarized as follows. In SMHG, the number of graph heads is set to $n = 5$. For dimensionality reduction, PCA retains $B = 30$ components before graph construction. For superpixel generation, the SLIC superpixel size parameter is set to 25 for all datasets, and the resulting number of superpixels varies with the image size of each dataset.

Regarding dataset configuration, for the IP and AG datasets, we randomly selected 1% of the labeled samples for training and an additional 1% for validation. In contrast, due to the abundance of labeled samples in the LK and PU datasets, we adopted a more stringent sampling strategy, using only 0.1% of the samples for training and another 0.1% for validation. During model optimization, we used the Adam optimizer with an initial learning rate of 0.001 and trained the model for 350 epochs to optimize the parameters. Table I summarizes the land-cover classes present in each of the four datasets, along with the corresponding numbers of samples allocated to the training, validation, and test sets.

To evaluate the classification performance of MGL-MKCNet, we adopt four widely used metrics: per-class accuracy (PA [%]), overall accuracy (OA [%]), average accuracy (AA [%]), and the Kappa coefficient ($\kappa \times 100$). In addition to presenting quantitative results in tabular form, we also generate visualizations of the classification maps produced by each competing method, enabling an intuitive qualitative

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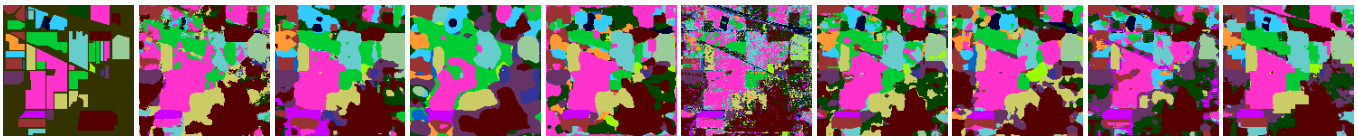


Fig. 4. Classification maps for different methods on the IP dataset. (a) Ground truth. (b) CEGCN. (c) WFCG. (d) FDGC. (e) SSFTT. (f) DSGSF. (g) GSC-ViT. (h) MambaHSI. (i) S2VNet. (j) Ours.

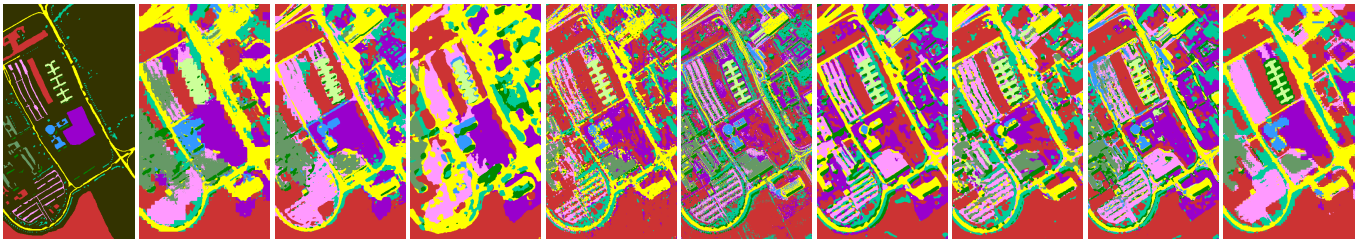


Fig. 5. Classification maps for different methods on the PU dataset. (a) Ground truth. (b) CEGCN. (c) WFCG. (d) FDGC. (e) SSFTT. (f) DSGSF. (g) GSC-ViT. (h) MambaHSI. (i) S2VNet. (j) Ours.

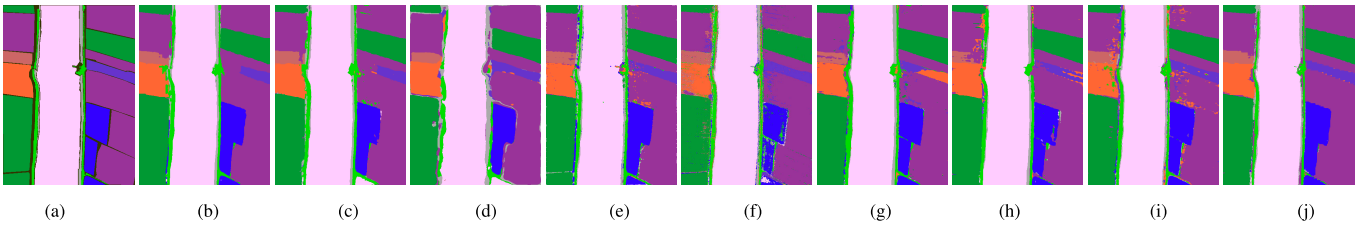


Fig. 6. Classification maps for different methods on the LK dataset. (a) Ground truth. (b) CEGCN. (c) WFCG. (d) FDGC. (e) SSFTT. (f) DSGSF. (g) GSC-ViT. (h) MambaHSI. (i) S2VNet. (j) Ours.

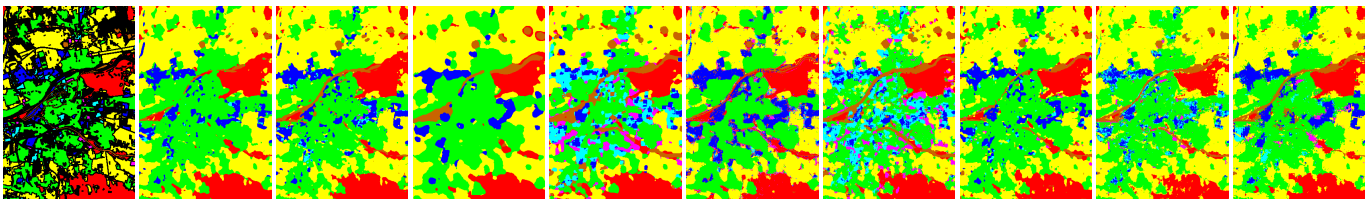


Fig. 7. Classification maps for different methods on the AG dataset. (a) Ground truth. (b) CEGCN. (c) WFCG. (d) FDGC. (e) SSFTT. (f) DSGSF. (g) GSC-ViT. (h) MambaHSI. (i) S2VNet. (j) Ours.

comparison. Moreover, all reported results are averaged over ten independent experimental runs to ensure statistical reliability and facilitate fair, precise comparisons.

B. Experimental Results

The classification accuracy of each model across the four datasets is reported in Tables II–V, while the corresponding classification maps produced by the various methods are visualized in Figs. 4–7.

1) IP Dataset: Although the IP dataset suffers from a limited number of labeled samples, our proposed MGL-MKCNet demonstrates superior classification performance compared to the state-of-the-art methods, as evidenced by the results in Table II and the classification map shown in Fig. 4. Specifically, MGL-MKCNet achieves an overall accuracy (OA) of 90.24%, an average accuracy (AA) of 86.33%, and a $\kappa \times 100$ of 88.86, outperforming all baseline models. Transformer-based models, SSFTT and GSC-ViT, also achieve relatively good overall performance, with SSFTT attaining an OA of 87.74% and GSC-ViT reaching 80.94%. However, SSFTT fails on Class 16, with 0% accuracy, likely due to the lack of sufficient training samples and the inability of SA mechanisms to effectively model rare classes. Nevertheless, the global attention mechanism enables them to capture long-range pixel relationships, contributing to higher OA despite localized failures. For graph-based methods, CEGCN and FDGC perform relatively poorly, which may be due to their limited ability to adaptively learn node-level interactions. In contrast, WFCG, which combines CNN with graph attention network, has achieved encouraging results, performing particularly well across multiple categories. This improvement can be attributed to the graph attention network ability to dynamically allocate attention weights between neighboring pixels, thereby

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TABLE II
CLASSIFICATION ACCURACY ON THE IP DATASET. BEST RESULTS ARE IN BOLD

Class	CEGCN	WFCG	FDGC	SSFTT	DSGSF	GSC-VIT	MambaHSI	S2VNet	Ours
1	19.79±16.37	90.08±5.99	62.59±25.84	95.71±6.61	54.03±25.74	63.56±17.02	95.71±4.36	37.73±19.59	**98.53±2.64**
2	78.20±8.80	**92.69±2.05**	80.86±6.30	80.34±4.91	59.37±15.19	74.21±10.16	79.34±7.04	66.70±23.49	86.35±5.91
3	46.37±13.74	84.31±10.36	74.07±12.69	**85.42±4.29**	58.97±19.81	70.81±2.60	69.30±7.34	74.66±16.60	83.11±2.61
4	35.03±21.21	81.47±6.59	69.13±8.63	81.15±9.49	48.34±15.89	58.63±14.31	63.92±17.06	56.11±28.64	**88.33±9.20**
5	74.92±8.97	**99.43±1.07**	84.02±11.44	90.19± 4.21	78.57±12.90	74.43±7.04	74.36±10.51	84.32±4.42	95.98±2.47
6	92.84±5.95	**99.33±0.64**	95.64±3.10	94.65±4.84	90.83±8.03	95.59±1.62	92.25±3.39	94.62±4.65	98.64±1.32
7	74.81±13.71	46.27±9.61	72.62±21.53	70.45±21.53	41.84±20.52	91.85±10.84	**100.00±0.00**	80.77±28.52	32.84±7.28
8	99.18±1.15	95.54±1.91	99.49±1.43	95.44±3.84	97.59±2.36	95.86±8.27	94.13±7.25	97.82±1.60	**100.00±0.00**
9	12.63±8.81	69.01±12.09	42.18±21.71	92.33±14.51	25.65±14.34	73.68±13.72	**97.50±5.73**	72.22±33.10	59.67±27.08
10	74.00±3.59	**90.76±4.19**	65.77±4.30	86.09±3.42	61.31±15.05	84.30±6.05	71.30±8.48	77.90±20.65	87.15±6.27
11	81.27±6.85	85.03±2.27	81.65±5.66	90.78±3.96	71.50±14.03	83.98±3.63	85.00±4.24	63.46±22.15	**93.37±2.56**
12	50.73±9.71	66.21±11.27	65.52±10.50	71.1±3.45	52.91±18.04	54.26±4.53	53.62±9.92	56.32±24.01	**79.06±7.12**
13	87.45±11.10	95.52±1.35	95.90±6.30	95.95±2.82	86.85±10.39	93.13±8.13	**99.20±0.75**	91.36±8.37	94.51±7.18
14	97.04±1.84	**98.28±1.53**	94.77±3.31	96.04±3.22	89.08±8.99	94.56±3.87	96.09±3.16	94.35±3.52	97.35±2.10
15	78.78±6.01	**90.64±2.78**	58.81±19.35	85.65±6.09	65.07±24.41	63.38±16.26	80.34±12.23	82.43±3.97	87.45±11.24
16	15.16±20.96	95.81±6.62	73.24±8.86	0.00±0.00	72.34±22.69	95.87±2.78	97.64±2.38	85.28±10.79	**98.98±1.65**
OA (%)	76.82±1.21	88.93±1.27	79.45±1.38	87.74±0.36	69.77±12.20	80.94±0.49	81.46±1.71	75.64±6.69	**90.24±0.93**
AA (%)	63.64±2.26	86.27±1.55	76.02±3.57	81.96±2.04	65.89±11.26	79.26±2.23	84.36±2.80	76.00±8.90	**86.33±2.19**
$\kappa \times 100$	73.46±1.32	87.34±1.45	76.52±1.58	86.02±0.43	65.43±14.03	78.22±0.55	78.05±3.96	72.47±7.54	**88.86±1.06**

TABLE III
CLASSIFICATION ACCURACY ON THE PU DATASET. BEST RESULTS ARE IN BOLD

Class	CEGCN	WFCG	FDGC	SSFTT	DSGSF	GSC-VIT	MambaHSI	S2VNet	Ours
1	91.25±8.96	**95.62±2.23**	75.12±4.90	79.56±3.24	94.17±2.14	93.99±2.87	78.92±11.72	83.50±7.91	92.61±6.07
2	97.30±1.45	97.30±0.98	92.05±1.89	94.46±5.51	89.46±3.45	88.59±8.01	94.58±3.25	**97.89±1.84**	97.68±0.81
3	85.71±12.30	78.52±11.27	76.16±13.82	81.87±8.88	56.46±14.41	68.56±10.51	59.71±15.87	63.30±25.78	**90.50±5.68**
4	77.86±1.97	83.87±4.01	89.13±5.04	82.80±10.38	90.76±7.94	80.41±8.82	61.79±14.06	84.32±11.36	**95.95±3.82**
5	**100.00±0.00**	99.87±0.26	89.72±3.60	99.82±0.36	97.48±2.34	99.97±0.06	98.47±2.84	99.54±0.43	92.37±7.53
6	**94.00±8.69**	88.42±1.59	93.35±9.59	83.65±13.41	57.93±11.56	78.65±7.32	83.96±12.81	63.51±22.44	92.37±7.53
7	88.91±11.04	**99.99±0.03**	85.49±8.56	96.74±1.74	57.59±17.10	38.58±35.91	70.81±21.08	88.27±6.07	97.43±1.75
8	85.37±9.53	**98.83±1.21**	58.12±6.99	47.03±12.84	81.01±6.60	57.22±19.11	75.18±11.34	80.84±14.35	78.41±8.39
9	57.83±21.85	59.89±2.29	60.22±25.90	0.00±0.00	**97.70±0.28**	94.40±3.12	92.45±6.34	93.16±5.35	89.11±15.55
OA (%)	91.94±1.66	93.58±0.35	83.25±1.94	83.49±0.56	81.22±4.03	82.92±4.72	84.51±2.09	87.13±2.84	**93.97±0.60**
AA (%)	86.47±3.55	89.15±1.31	79.93±3.61	73.99±1.61	80.29±4.14	77.82±3.32	79.54±2.26	83.81±3.21	**92.58±0.93**
$\kappa \times 100$	89.29±2.24	91.44±0.47	77.48±2.53	78.12±0.68	75.44±5.23	77.61±5.72	75.91±8.47	82.70±3.89	**91.97±0.80**

TABLE IV
CLASSIFICATION ACCURACY ON THE LK DATASET. BEST RESULTS ARE IN BOLD

Class	CEGCN	WFCG	FDGC	SSFTT	DSGSF	GSC-VIT	MambaHSI	S2VNet	Ours
1	**99.85±0.13**	99.77±0.07	97.44±1.43	98.92±0.55	98.17±1.58	99.53±0.25	99.25±0.64	99.64±0.35	99.70±0.18
2	99.85±0.13	**99.88±0.16**	91.88±6.49	97.27±1.27	90.96±2.04	91.67±4.85	92.46±3.26	97.23±0.88	97.01±1.64
3	80.61±13.06	88.84±1.75	89.31±10.01	81.28±10.19	76.12±9.15	66.57±19.44	71.03±12.05	87.37±9.58	**96.07±5.60**
4	**99.48±0.23**	99.22±0.25	97.32±1.79	97.20±1.25	97.86±1.12	98.02±0.81	97.97±0.69	99.15±0.55	98.48±0.73
5	87.77±14.17	**95.04±2.67**	70.09±1.75	73.33±3.71	74.68±15.43	75.66±14.01	62.55±17.95	83.35±10.70	91.74±3.43
6	99.39±0.36	93.80±1.20	97.18±1.38	91.27±4.00	95.99±3.90	91.97±2.18	97.74±1.68	98.65±0.63	**99.60±0.41**
7	99.99±0.01	**100.00±0.00**	99.83±0.16	99.94±0.09	99.02±0.65	99.70±0.55	99.89±0.21	99.71±0.34	99.52±0.64
8	95.95±1.20	96.91±1.49	76.58±3.92	89.08±4.76	90.88±4.64	79.31±10.99	79.04±9.27	90.40±9.42	**99.52±0.64**
9	64.86±6.02	75.00±1.42	65.38±12.36	79.16±7.60	**92.00±3.09**	73.57±12.56	77.00±9.80	82.07±9.57	91.88±3.60
OA (%)	97.87±0.33	98.34±0.07	95.30±0.91	96.58±0.36	96.50±0.66	96.02±1.00	96.26±0.74	98.07±0.41	**98.52±0.28**
AA (%)	91.10±1.71	94.27±0.21	87.22±1.15	89.72±1.46	90.63±1.76	86.22±3.87	86.32±3.66	93.06±1.54	**96.47±0.86**
$\kappa \times 100$	97.19±0.44	97.81±0.09	93.84±1.20	95.50±0.46	95.40±0.87	94.75±1.33	94.30±2.08	97.47±0.54	**98.06±0.37**

capturing complex spatial dependencies and enhancing feature representations.

As shown in Fig. 4, all comparison methods except WFCG yield poor classification performance on few-sample classes. The classification results of S2VNet and DSGSF exhibit distinct "salt-and-pepper" noise, indicating poor spatial consistency and unstable predictions. In contrast, the classification map generated by MGL-MKCNet is smoother and exhibits stronger spatial consistency, suggesting that it is highly capable of modeling both local and global contextual information. In general, the experimental results demonstrate that MGL-MKCNet not only achieves the highest OA, AA, and $\kappa \times 100$ values, but also generates visually coherent and semantically accurate classification maps, confirming its robustness and effectiveness in handling challenging HSI classification tasks with limited labeled data.

2) PU Dataset: The PU dataset presents a challenging classification task due to its high spectral complexity and spatial heterogeneity, as shown in Table III. Our proposed MGL-MKCNet achieves the best overall performance across all evaluation metrics: an OA of 93.97%, an AA of 92.58%, and a $\kappa \times 100$ of 91.97. These results indicate superior generalization ability and robust feature representation under complex urban environments. Among the compared methods, WFCG also achieves strong results (OA: 93.58%, AA: 89.15%, $\kappa \times 100$: 91.44) by leveraging graph attention network

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TABLE V
CLASSIFICATION ACCURACY ON THE AG DATASET. BEST RESULTS ARE IN BOLD

Class	CEGCN	WFCG	FDGC	SSFTT	DSGSF	GSC-VIT	MambaHSI	S2VNet	Ours
1	**99.10±0.49**	97.28±0.54	94.68±0.66	92.51±2.49	93.11±2.51	97.68±0.36	97.60±0.47	97.60±2.14	97.18±0.53
2	**99.10±0.27**	96.68±0.59	94.62±1.67	97.79±1.14	93.05±3.42	83.50±3.14	97.74±0.66	95.98±1.41	95.67±0.67
3	75.04±4.37	**75.64±4.47**	69.95±9.11	63.93±6.65	69.64±8.51	37.29±17.13	71.39±9.14	64.22±5.19	74.58±3.29
4	**98.97±0.28**	98.41±0.24	95.34±0.94	97.93±1.08	94.95±1.12	86.12±3.91	96.08±0.78	98.23±1.92	98.25±0.37
5	0.11±0.21	26.18±33.50	9.23±13.74	32.78±20.26	40.34±12.69	**89.98±4.57**	12.48±12.05	59.68±11.85	70.28±9.86
6	13.81±6.19	30.45±13.34	26.37±17.01	30.77±11.76	35.03±10.86	**71.43±10.56**	30.8±12.13	40.25±9.85	53.78±8.17
7	40.85±4.28	66.45±8.71	46.23±5.00	29.69±15.75	78.01±13.20	69.65±2.09	41.26±7.81	56.16±5.01	**85.29±4.20**
OA (%)	94.21±0.24	94.52±0.23	91.60±1.26	84.31±5.06	91.28±1.80	84.12±1.76	92.71±0.51	93.25±0.41	**94.89±0.14**
AA (%)	61.00±1.42	70.14±6.85	62.35±5.65	63.63±3.19	72.02±3.52	76.52±2.35	63.91±3.32	73.16±2.19	**82.15±2.10**
$\kappa \times 100$	91.59±0.37	87.81±0.31	77.70±1.98	87.45±6.20	87.45±2.63	78.28±2.26	90.09±0.91	90.34±0.55	**92.63±0.21**

adaptive attention to enhance local feature learning. CEGCN (OA: 91.94%, AA: 86.47%, $\kappa \times 100$: 89.29) significantly outperforms FDGC, suggesting that GCNs are better suited than general GNNs for modeling graph-structured features in dense urban scenes. S2VNet performs notably well (OA: 87.13%, AA: 83.81%, $\kappa \times 100$: 82.70), likely thanks to its deep architecture that captures subpixel-level spectral–spatial variations. In contrast, GSC-ViT underperforms (OA: 82.92%, AA: 77.82%, $\kappa \times 100$: 77.61), possibly due to its difficulty in resolving fine spatial details despite its global attention mechanism.

Furthermore, as illustrated in Fig. 5, the classification map generated by MGL-MKCNet exhibits excellent noise suppression and high spatial consistency. Particularly noteworthy is the model's ability to clearly delineate class regions, especially for large-scale features, where its internal classification accuracy surpasses that of other models. This visual evidence further confirms that the model effectively captures local texture and global contextual information, thereby achieving semantically accurate and visually consistent classification results.

3) WHU-Hi-LongKou (LK) Dataset: As shown in Table IV, WFCG achieved the highest accuracy across multiple categories and ranked second overall, indicating that the model performs exceptionally well on this large-scale dataset. S2VNet also demonstrated significant improvements, suggesting that it is well-suited to high-resolution HSI classification tasks with extensive spatial coverage. In contrast, FDGC performed poorly, indicating that it is limited in its ability to capture complex spectral–spatial patterns.

Our proposed MGL-MKCNet achieves the best results across all metrics: OA = 98.51%, AA = 96.69%, and $\kappa \times 100$ = 98.04, outperforming all baseline models. As illustrated in Fig. 6, it produces the cleanest classification map with minimal noise and sharp boundaries, especially for small-scale features, further validating its superior robustness and spatial fidelity.

4) Augsburg (AG) Dataset: As shown in Table V, our proposed model continues to achieve optimal performance compared to other models, with an OA of 94.89%, an AA of 82.15%, and a $\kappa \times 100$ of 92.63. This demonstrates that our proposed model possesses excellent generalization capabilities, delivering outstanding performance on both large datasets and smaller datasets. Similarly, we observe that the Transformer-based SSFTT and GSC-VIT models did not yield favorable results. This may stem from SSFTT's shallow convolutions struggling to capture complex spatial context, while feature serialization disrupts hyperspectral three-dimensional structures, leading to loss of critical spectral–spatial correlation information. Conversely, GSC-VIT's lightweight grouped separable convolutions weaken its high-dimensional feature extraction capabilities, and its traditional multihead attention mechanism amplifies interference from redundant spectral band information. Furthermore, the Transformer architecture itself exhibits limited capability in capturing fine-grained local features, struggling to adapt to the spatial demands of the dataset, ultimately resulting in suboptimal classification performance. Conversely, CNN-based models, such as CEGCN and WFCG, achieved superior results, primarily due to the CNN architecture's robust ability to capture local spatial features.

The visualization results in Fig. 7 clearly demonstrate that our proposed model exhibits the lowest noise interference, whilst rendering land feature edges with greater clarity and distinctiveness. This outcome further validates the effectiveness of the proposed model in HSI classification tasks.

C. Influence of Training Samples

To investigate the comparative performance of MGL-MKCNet against other models under varying sample sizes, four datasets were selected for experimentation. The IP and AG datasets featured training samples ranging from 1% to 5% per class. Conversely, the PU and LK datasets exhibited training sample proportions between 0.1% and 0.5% per class. The comparative metric employed was OA (%). As illustrated in Fig. 8, MGL-MKCNet achieved the highest classification accuracy across all four datasets.

From the line charts corresponding to four distinct datasets, it is evident that as the number of training samples continues to increase, the performance of the vast majority of models exhibits a steady upward trend. Similarly, we observe a peculiar behavior in the FDGC model: when the sample size is substantial, its performance undergoes significant fluctuations. This phenomenon is most pronounced in Fig. 8(c), which corresponds to the LK dataset. We hypothesize that this fluctuation is likely caused by overfitting within the FDGC model, resulting from the excessive scale of the sample size.

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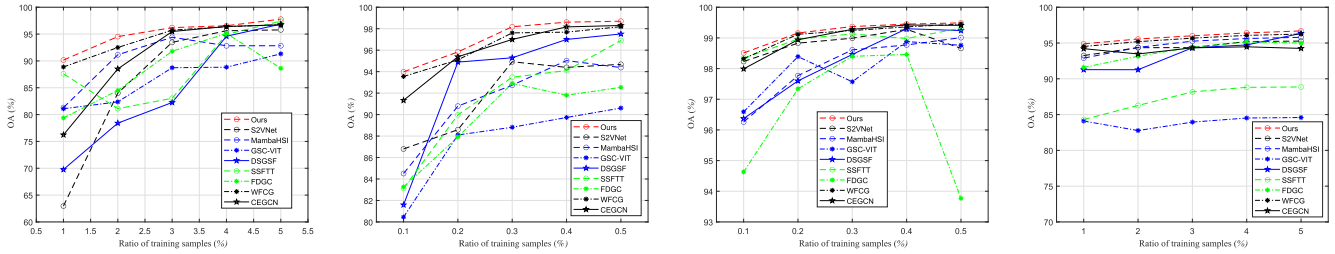


Fig. 8. Effect of different training sample proportions on the classification performance on different datasets. (a) IP. (b) PU. (c) LK. (d) AG.

TABLE VI

ABLATION STUDIES OF SMHG, SSCA, AND MKCONV ON THE FOUR HSI DATASETS. THE EVALUATION METRIC IS OA (%)

SMHG	SSCA	MKConv	IP	PU	LK	AG
✓	✗	✗	85.93	89.75	96.73	91.53
✗	✓	✗	56.12	72.54	89.06	88.22
✗	✗	✓	64.62	73.07	79.11	90.47
✓	✓	✗	86.58	90.08	96.96	92.08
✓	✗	✓	88.75	91.95	98.35	96.73
✗	✓	✓	65.21	76.93	91.40	94.35
✓	✓	✓	**90.16**	**93.98**	**98.51**	**94.89**

TABLE VII

MODEL PERFORMANCE UNDER VARYING NUMBERS OF GRAPH LAYERS IN SMHG ON THE IP DATASET

Number	OA (%)	AA (%)	$\kappa \times 100$
1	88.40	83.32	86.79
2	88.29	84.43	86.66
3	89.00	85.48	87.43
4	90.09	86.77	88.68
5	**90.38**	**87.42**	**89.04**
6	88.96	84.56	87.40

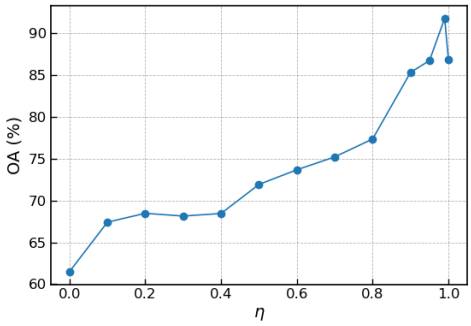


Fig. 9. OA of the proposed MGL-MKCNet on the IP dataset under different values of η.

D. Ablation Experiments

To validate the contribution of each proposed component, we conducted a series of ablation studies on the IP, PU, LK, and AG datasets. Specifically, we evaluated variants of the model incorporating: 1) only the SMHG module; 2) only the SSCA module; 3) only the MKConv module; 4) combinations of any two modules; and 5) the full model integrating all three modules. All results were averaged over ten independent runs, with OA used as the evaluation metric.

As shown in Table VI, the ablation results not only verify the effectiveness of each module, but also support the underlying mechanism of MGL-MKCNet. SMHG provides region-level global contextual modeling, SSCA enhances spectral–spatial interaction before nonlinear transformation, and MKConv further refines pixel-level features through adaptive multiscale nonlinear modeling. The substantial performance gaps between the full model and single-module settings further highlight the critical role of module integration in achieving robust HSI classification.

E. Impact of Different Numbers of Graph Layers

As shown in Table VII, the experimental results reveal a clear trend in model performance: it gradually improves with an increase in the number of graph layers, reaching a peak at five layers. Beyond this point, further increases in the number of graph layers yield diminishing returns. Consequently, setting the number of graph layers to five achieves optimal performance across all metrics.

F. Impact of Different Fusion Coefficients

As illustrated in Fig. 9, the OA gradually improves as η increases and reaches its maximum around $\eta = 0.99$. This trend indicates that, in the current framework, the graph-based branch contributes more strongly to the final prediction, since it captures global topological relationships and contextual dependencies among superpixels. However, when $\eta = 1$, the performance drops noticeably. This suggests that relying solely on the graph-based branch is suboptimal, because the pixel-based branch still provides complementary fine-grained local spectral–spatial details that are beneficial for accurate classification. Therefore, $\eta = 0.99$ achieves the best tradeoff in our model by assigning dominant weight to the global contextual features while retaining a small but useful contribution from the pixel-level branch. It should be noted that this value of $\eta = 0.99$ represents the empirically optimal setting under the current experimental conditions, rather than a universally applicable constant.

G. Comparison Between FastKAN and Standard KAN

To validate the rationale for selecting FastKAN in the proposed MKConv module, we conducted a small-scale comparative experiment between FastKAN and standard KAN

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TABLE VIII

COMPARISON BETWEEN FASTKAN AND STANDARD KAN. THE BEST RESULT IS SHOWN IN BOLD

Variant	OA (%)	AA (%)	$\kappa \times 100$	Time (s)
Standard KAN	87.55	85.38	85.88	11.67
FastKAN	**90.24**	**86.33**	**88.86**	**9.84**

TABLE IX

COMPUTATIONAL EFFICIENCY COMPARISON OF DIFFERENT MODELS ON THE IP DATASET. THE BEST RESULT IS SHOWN IN BOLD

Metrics	CEGCN	WFCG	FDGC	SSFTT	DSGSF	GSC-VIT	MambaHSI	S2VNet	Ours
Params (M)	0.17	**0.10**	2.45	0.15	0.44	0.64	0.43	0.23	0.33
Training (s)	3.70	5.74	2.12	**2.09**	29.68	5.88	36.27	125.80	9.84
Test (s)	0.17	0.13	0.88	0.23	1.06	1.36	**0.04**	1.76	0.17

on the IP dataset. The training set was fixed at 1% of the total sample size, the validation set was also set at 1%, and the remaining samples were allocated to the test set. Performance evaluation metrics included OA, AA, $\kappa \times 100$, and training time (s). Specifically, while keeping the rest of the MGL-MKCNet architecture unchanged, we replaced the FastKAN-based convolutions in MKConv with standard KAN convolutions, and subsequently compared their classification performance and computational overhead.

As shown in Table VIII, FastKAN consistently outperforms the standard KAN across all evaluation metrics. Specifically, FastKAN achieves an OA of 90.24%, AA of 86.33%, and $\kappa \times 100$ of 88.86, which are all significantly higher than those obtained by standard KAN (87.55%, 85.38%, and $\kappa \times 100$ of 85.88, respectively). It also outperforms the standard KAN in terms of training time.

This performance improvement demonstrates that FastKAN provides a more effective nonlinear representation capability within the MKConv module. Moreover, FastKAN avoids the instability and computational overhead associated with B-spline-based KAN, making it more suitable for HSI classification tasks.

H. Computational Complexity Analysis of Different Models

We evaluate the computational efficiency of the proposed method and other baseline models on the IP dataset under a unified experimental setting, where 1% labeled samples are used for training, 1% for validation, and the remaining samples for testing. Three core metrics, including model parameters (Params, M), training time (s), and test time (s), are adopted for quantitative comparison, and the optimal results of each metric are marked in bold in Table IX.

In terms of parameter occupancy, WFCG obtains the smallest parameter size of 0.10M, followed by SSFTT (0.15M) and CEGCN (0.17M). By contrast, FDGC requires massive parameters up to 2.45M, resulting in a heavy memory burden. Our model only maintains a moderate parameter quantity of 0.33M, which well balances feature representation capability and memory consumption without exceeding the common threshold of mainstream lightweight models. In terms of training efficiency, SSFTT achieves the fastest training speed of 2.09 s. Compared with time-consuming complex models such as S2VNet and MambaHSI, our method delivers a competitive training efficiency of 9.84 s, which obviously outperforms advanced transformer-based and Mamba-based competitors. For inference efficiency, MambaHSI ranks first with a test latency of 0.04 s. Importantly, our model achieves a comparable low test time of 0.17 s, which is consistent with lightweight GCN-based models and much faster than other competitive methods, including DSGSF and GSC-VIT. In conclusion, the proposed method achieves an excellent balance between classification performance and computational overhead. It possesses moderate model scale, high training efficiency, and fast inference speed, effectively overcoming the common contradiction between accuracy and deployment cost in HSI classification.

V. CONCLUSION

We propose an MGL-MKCNet for HSI classification. The proposed architecture unifies contextual modeling based on global graphs, adaptive learnable nonlinearity through the KAN, and SSCA within an end-to-end framework, thus synergistically integrating the advantages of CNNs and GCNs while overcoming their inherent limitations. Specifically, we design an SSCA module to utilize spectral–spatial information efficiently. Based on the FastKAN, we transform traditional CNNs into an MKConv module. Simultaneously, drawing inspiration from multihead attention mechanisms, we propose an SMHG module. By integrating multiple feature streams, this module better learns features from numerous neighbors. In the experiments, the model was evaluated on four HSI datasets. Compared with eight state-of-the-art methods, it performs optimally in three metrics. Ablation experiments demonstrate that the SMHG, SSCA, and MKConv modules each improve model performance, with their combination yielding the best results. Future work will focus on the following directions: optimizing computational efficiency by proposing lightweight models to handle larger HSI datasets and attempting to apply these models to other HSI processing tasks, such as spectral unmixing, change detection, and target detection. Furthermore, efforts will be made to enhance the models' generalization capabilities by incorporating multimodal data.

REFERENCES

[1] Z. Ye, H. Li, Y. Song, J. A. Benediktsson, and Y. Y. Tang, "Hyperspectral image classification using principal components-based smooth ordering and multiple 1-D interpolation," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 2, pp. 1199–1209, Feb. 2017.

[2] S. Peyghambari and Y. Zhang, "Hyperspectral remote sensing in lithological mapping, mineral exploration, and environmental geology: An updated review," *J. Appl. Remote Sens.*, vol. 15, no. 3, Jul. 2021, Art. no. 031501.

[3] Y. Zhang, Z. Ye, C. Zheng, B. Yang, J. Peng, and W. Sun, "Hyperspectral marine oil spill detection network with enhanced superpixel segmentation and attention mechanisms," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4204213.

[4] Y. Wang, S. Mei, M. Ma, Y. Liu, T. Gao, and H. Han, "Hyperspectral object tracking with context-aware learning and category consistency," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5508716.

[5] Y. Su, L. Gao, A. Plaza, X. Sun, M. Jiang, and G. Yang, "SRViT: Self-supervised relation-aware vision transformer for hyperspectral unmixing," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 17585–17598, Oct. 2025.

[6] Z. Ye, L. Zhang, J. Peng, J. A. Benediktsson, and T. Qian, "Orthogonal rational basis-function-based discriminant analysis for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, 2026, Art. no. 5514118.

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[7] J. Peng, L. Li, and Y. Y. Tang, "Maximum likelihood estimation-based joint sparse representation for the classification of hyperspectral remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 6, pp. 1790–1802, Jun. 2019.

[8] C. Zheng, Z. Ye, J. Cui, and J. Peng, "Two-stage hyperspectral image classification using few labeled samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, 2024, Art. no. 5504105.

[9] Y. Su et al., "Dilated transformation-guided unsupervised multimodal learning for hyperspectral and multispectral image fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5535715.

[10] Y. Huang, J. Peng, W. Sun, N. Chen, Z. Ye, and Q. Du, "Domain-aware adversarial domain augmentation network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 35, pp. 1275–1289, 2026.

[11] Q. Liu, J. Peng, G. Zhang, W. Sun, and Q. Du, "Deep contrastive learning network for small-sample hyperspectral image classification," *J. Remote Sens.*, vol. 3, p. 25, Jan. 2023.

[12] F. Xu, S. Mei, G. Zhang, N. Wang, and Q. Du, "Bridging CNN and transformer with cross-attention fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5522214.

[13] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[14] Z. Zhao, X. Xu, S. Li, and A. Plaza, "Hyperspectral image classification using groupwise separable convolutional vision transformer network," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5511817.

[15] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216.

[16] B. Yang, S. Pan, W. Sun, Z. Guo, Z. Ye, and J. Peng, "FCFDA: Fine-coarse-fine progressive graph framework with distribution alignment for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538714.

[17] Y. Lu, S. Mei, F. Xu, M. Ma, and X. Wang, "Separable deep graph convolutional network integrated with CNN and prototype learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516216.

[18] Y. Chen, K. Zhu, L. Zhu, X. He, P. Ghamisi, and J. A. Benediktsson, "Automatic design of convolutional neural network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 7048–7066, Sep. 2019.

[19] S. Mei, X. Chen, Y. Zhang, J. Li, and A. Plaza, "Accelerating convolutional neural network-based hyperspectral image classification by step activation quantization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502012.

[20] X. Liu, A. H.-M. Ng, F. Lei, J. Ren, L. Guo, and Z. Du, "MSLKCNN: A simple and powerful multiscale large kernel CNN for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5513314.

[21] S. Zhang, M. Xu, J. Zhou, and S. Jia, "Unsupervised spatial–spectral CNN-based feature learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5524617.

[22] G. Wu, M. A. A. Al-Qaness, D. Al-Alimi, A. Dahou, M. A. Elaziz, and A. A. Ewees, "Hyperspectral image classification using graph convolutional network: A comprehensive review," *Expert Syst. Appl.*, vol. 257, Dec. 2024, Art. no. 125106.

[23] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[24] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel- and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[25] Q. Liu, Y. Dong, Y. Zhang, and H. Luo, "A fast dynamic graph convolutional network and CNN parallel network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530215.

[26] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.

[27] Z. Liu et al., "KAN: Kolmogorov–Arnold networks," 2024, *arXiv:2404.19756*.

[28] F. Zhang and X. Zhang, "GraphKAN: Enhancing feature extraction with graph Kolmogorov Arnold networks," 2024, *arXiv:2406.13597*.

[29] Z. Li, "Kolmogorov–Arnold networks are radial basis function networks," 2024, *arXiv:2405.06721*.

[30] S. Zhang, H. Tong, J. Xu, and R. Maciejewski, "Graph convolutional networks: A comprehensive review," *Comput. Social Netw.*, vol. 6, no. 1, pp. 1–23, Dec. 2019.

[31] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Proc. Adv. Neural Inf. Process. Syst.*, 2016, pp. 1–9.

[32] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," 2016, *arXiv:1609.02907*.

[33] A. D. Bodner, A. S. Tepsich, J. N. Spolski, and S. Pourteau, "Convolutional Kolmogorov–Arnold networks," 2024, *arXiv:2406.13155*.

[34] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Süsstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[35] Y. Si et al., "SCSA: Exploring the synergistic effects between spatial and channel attention," *Neurocomputing*, vol. 634, Mar. 2025, Art. no. 129866.

[36] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 1251–1258.

[37] T. Guo, R. Wang, F. Luo, X. Gong, L. Zhang, and X. Gao, "Dual-view spectral and global spatial feature fusion network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5512913.

[38] Z. Han, J. Yang, L. Gao, Z. Zeng, B. Zhang, and J. Chanussot, "Subpixel spectral variability network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5504014.

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